Solve with us 9 - Not Graded

This assignment will not be graded and is only for practice.
Key Points: 1Key Points: 1

- Scalar-valued multivariable functions:Scalar-valued multivariable functions: A scalar-valued multivariable function is a function $f : D \to R$f : D → R, where DD is a domain in R^n Rn where $n > 1$n > 1.
- Vector-valued multivariable functions:Vector-valued multivariable functions: A vector-valued multivariable function is a function $f : D \to R^m$ f : D → Rm, where DD is a domain in R^n Rn where $n, m > 1$n, m > 1.

1 point
Which of the following functions are scalar-valued multivariable functions?

☐ $f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).

☐ $f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).

☐ $f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).

☐ $f(x, y) = (xy, 0)$f(x, y) = (xy, 0).

☐ $f(x, y) = xy$f(x, y) = xy.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).
$f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).
$f(x, y) = xy$f(x, y) = xy.

1 point
Which of the following functions are vector-valued multivariable functions?

☐ $f(x, y) = xy + sin(xy)$f(x, y) = xy + sin(xy).

☐ $f(x, y, z) = ln(x + y + z)$f(x, y, z) = ln(x + y + z).

☐ $f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).

☐ $f(x, y) = (xy, 0)$f(x, y) = (xy, 0).

☐ $f(x, y) = xy$f(x, y) = xy.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y, z) = (x^2, z, y + xz)$f(x, y, z) = (x2, z, y + xz).
$f(x, y) = (xy, 0)$f(x, y) = (xy, 0).

Key Points: 2Key Points: 2

- Composition of functions:Composition of functions: Let $D \subseteq R^n$ and $f: D \to R^m$ be a multivariable function. Let $g: E \to R^p$ be a function on E where $Range(f) \subseteq E \subseteq R^m$.

  We define a multivariable function $g \circ f: D \to R^p$ called the composition of f and g defined as $g \circ f(\tilde{x}) = g(f(\tilde{x}))$, for all $\tilde{x} \in D$.

1 point

If two functions $f: R^2 \to R$ and $g: R \to R$ are defined as $f(x, y) = (x - y)$ and $g(x) = \sin x$, then which of the following functions represents $g \circ f$?

- ◯ $\sin x - \sin y$.
- ◯ $\sin(x - y)$.
- ◯ $x \sin x - y \sin y$.
- ◯ $\sin xy$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\sin(x - y)$.

1 point

If two functions $f: R^2 \to R$ and $g: R \to R^2$ are defined as $f(x, y) = xy$ and $g(t) = (e^t, e^{-t})$, then which of the following functions represents $g \circ f$?

- ◯ $g \circ f(x, y) = e^{xy}$.
- ◯ $g \circ f(x, y) = 1$.
- ◯ $g \circ f(x, y) = (e^{xy}, e^{-xy})$.
- ◯ $g \circ f(x, y) = (e^x, e^{-x})$.
- ◯ $g \circ f(x, y) = (1, 1)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g \circ f(x, y) = (e^{xy}, e^{-xy})$.

Key Points: 3Key Points: 3

- Partial derivatives:Partial derivatives: Let $f(x_1, x_2, \ldots, x_n)$ be a function defined on a domain D in R^n. The partial derivative of f with respect to x_i is the function denoted by $f_{x_i}(\tilde{x})$ or $\frac{\partial f}{\partial x_i}(\tilde{x})$ and defined as

$$\frac{\partial f}{\partial x_i}(\tilde{x}) = \lim_{h \to 0} \frac{f(\tilde{x} + he_i) - f(\tilde{x})}{h}$$

where e_i is the unit vector $(0, 0, \ldots, 0, 1, 0, \ldots, 0)$ (at i-th coordinate there is 1, and 0 at all the other coordinates).

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f((x_1, x_2, \ldots, x_n) + h(0, 0, \ldots, 0, 1, 0, \ldots, 0)) - f(x_1, x_2, \ldots, x_n)}{h}$$

$$\frac{\partial f}{\partial x_i}(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{h}$$

- To calculate the partial derivative with respect to x_i, think of f only as a function of x_i while treating all other variables as constants. Then calculate it as the derivative of a function of one variable.

1 point

Consider the function $f : R^2 \to R$ such that $f(x, y) = x^2 + y^2$. Which of the following function will represent f_x or $\frac{\partial f}{\partial x}$?

○ $2x + 2y$.

○ $2x$.

○ $2x + y^2$.

○ $x + y^2$.

No, the answer is incorrect.
Score: 0
Feedback:

Treat y as constant.

Accepted Answers:

$2x$.

1 point

Consider the function $f : D \to R$ where $D \subset R^2$, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

What will be the partial derivative of f with respect to x at the point $(0, 0)$?

○ 1.

○ 0.

○ 2

○ Cannot be determined.

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
00.

Key Points: 4Key Points: 4

- Directional derivative:Directional derivative: The rate of change of $f(x_1, x_2, \ldots, x_n)$ in the direction of the unit vector $u = (u_1, u_2, \ldots, u_n)$ is called the directional derivative and is denoted by $D_u f(x_1, x_2, \ldots, x_n)$. The definition of the directional derivative is:

$$D_u f(x_1, x_2, \ldots, x_n) = \lim_{h \to 0} \frac{f(x_1 + u_1 h, x_2 + u_2 h, \ldots, x_n + u_n h) - f(x_1, x_2, \ldots, x_n)}{h}$$

- To calculate the directional derivative of a function f in a particular direction, we have to first find the unit vector u in that direction and then compute $D_u f$.
- The partial derivatives are special cases of the directional derivative where $u = e_i$ where e_i is i-th vector of the standard ordered basis of R^n.

1 point
Consider the function $f: R^2 \to R$, such that $f(x, y) = x + y$. Which of the following options will denote the directional derivative of f in the direction of the vector $v = (1, 1)$?

[Hint: The given vector v is not a unit vector. Convert v to a unit vector $u = \frac{v}{||v||}$.]

○ 2

○ $x + y$

○ $\sqrt{2}$

○ $\sqrt{2}(x + y)$

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
$\sqrt{2}$

Key Points: 5Key Points: 5

- Limit of a scalar valued multivariable function: If there exists some real number L such that $f(x_1, x_2, \ldots, x_n)$ approaches L as $(x_1, x_2, \ldots, x_n)$ approaches $(a_1, a_2, \ldots, a_n)$, then we write

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = L$$

- For functions of one variable $f(x)$, there are only two ways in which x can approach a point a, from left or from right. But for multivariable functions, $(a_1, a_2, \ldots, a_n)$ can be approached along any curve.
- If C is a curve passing through $(a_1, a_2, \ldots, a_n)$, the limit of f at $(a_1, a_2, \ldots, a_n)$ along the curve C exists and equals L if $f(x_1, x_2, \ldots, x_n)$ approaches L as $(x_1, x_2, \ldots, x_n)$ approaches $(a_1, a_2, \ldots, a_n)$ where $(x_1, x_2, \ldots, x_n)$ belongs to C.
- The limit of the function at $(a_1, a_2, \ldots, a_n)$ along the curve C equals L for each curve C passing through $(a_1, a_2, \ldots, a_n)$ precisely when the limit of f at $(a_1, a_2, \ldots, a_n)$ equals L.
- To show that the limit of f at $(a_1, a_2, \ldots, a_n)$ does not exist, one should either produce a curve C so that the limit of f at $(a_1, a_2, \ldots, a_n)$ along C does not exist or produce two curves so that the limits of f at $(a_1, a_2, \ldots, a_n)$ along both curves are not equal.
- Continuity of a function: If limit of a function $f: D \to R$, where $D \subseteq R^n$, exists at a point $(a_1, a_2, \ldots, a_n)$, and the functional value and the limiting value are the same, i.e.,

$$\lim_{(x_1, x_2, \ldots, x_n) \to (a_1, a_2, \ldots, a_n)} f(x_1, x_2, \ldots, x_n) = f(a_1, a_2, \ldots, a_n)$$

then we say f is continuous at the point $(a_1, a_2, \ldots, a_n)$.

Consider the function $f: D \to R$ where $D \subset R^2$, defined as:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^3 + y^3} & \text{if } (x, y) \neq (0,0) \\ 0 & \text{if } (x, y) = (0,0) \end{cases}$$

Answer the following question 8 to 13 using the above information.

If f approaches L as (x, y) approaches to the origin along the X-axis, then find the value of L.

No, the answer is incorrect.
Score: 0
Feedback:
Along the X-axis, y = 0.
Accepted Answers:

(Type: Numeric) 0

1 point

If f approaches L as (x, y) approaches to the origin along the Y -axis, then find the value of L.

No, the answer is incorrect.

Score: 0

Feedback:

Along the Y -axis, x = 0.

Accepted Answers:

(Type: Numeric) 0

1 point

If f approaches L as (x, y) approaches to the origin along the straight line $y = x$, then find the value of L.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

(Type: Numeric) 0.5

1 point

1 point

If f approaches L as (x, y) approaches to the origin along the straight line $y = 2x$, then find the value of L.

- 0.
- $\frac{1}{2}$.
- $\frac{2}{9}$.
- 1.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:

$\frac{2}{9}$.

1 point

If f approaches to L as (x, y) approaches to the origin along the straight line $y = mx$, then find the value of L.

- 0.
- $\frac{1}{m^2}$.
- $\frac{m}{1+m^3}$.
- 1.

No, the answer is incorrect.

Score: 0

Feedback:

Accepted Answers:
$\frac{m}{1+m^3}$.
1 point
Choose the set of correct options.

- [ ] $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 0$.
- [ ] f is not continuous at the origin.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = \frac{1}{2}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
f is not continuous at the origin.

1 point
Consider the function $f : D \to R$ where $D \subset R^2$, defined as:

$$f(x,y) = \begin{cases} \frac{y}{x^2+y} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Choose the set of correct options.
[Hint: Approach the origin using the curves $y = x^2$ and $y = 2x^2$.]

- [ ] $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 0$.
- [ ] f is not continuous at the origin.
- [ ] $\lim_{(x,y)\to(0,0)} f(x,y) = 1$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.
f is not continuous at the origin.

Key Points: 6

- Gradient of a function: If f is a scalar valued function defined on a domain D in R^n, then the gradient ∇f is defined as the vector valued function $\left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n}\right)$.
- If ∇f exists and is continuous in an open ball around the point $(a_1, a_2, \dots, a_n)$, then the directional derivative of the function f at $(a_1, a_2, \dots, a_n)$ in the direction of a vector v is given by the dot product of the gradient of the function at $(a_1, a_2, \dots, a_n)$ with the unit vector in the direction of v i.e.,

$$D_v f(a_1, a_2, \dots, a_n) = \nabla f(a_1, a_2, \dots, a_n) . \frac{v}{||v||}$$

1 point

Find the directional derivative of the function $f: R^2 \to R$ such that $f(x, y) = 3x^2 - 2xy$ in the direction of the vector $(3, 4)$.

- ○ $2x - \frac{6y}{5}$.
- ○ $10x - 6y$.
- ○ $10x$.
- ○ $2x$.
- ○ $-\frac{6y}{5}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$2x - \frac{6y}{5}$.

1 point

Find the directional derivative of the function $f: R^2 \to R$ such that $f(x, y) = 3x^2 + 2y^2$ at the point $(1, 1)$ along the direction of it's gradient vector $\nabla f(1, 1)$.
[Hint: First find out the unit vector in the direction of the vector $\nabla f(1, 1)$.]

- ○ 2
- ○ $\sqrt{13}$
- ○ $2\sqrt{13}$
- ○ $2\sqrt{2}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$2\sqrt{13}$